

RobotAPO: Adversarial Physics Preference Optimization for Robotic Manipulation Video Generation

Kerui Li^{1,2}, Zhe Jing³, Chenyi Huang^{1,2}, Xiaofeng Wang², Zheng Zhu², Haoming Cui^{1,2} and Huaibo Huang¹

¹Institute of Automation, Chinese Academy of Sciences, ²GigaAI, ³Beijing Institute of Technology

Abstract

Robotic manipulation videos are increasingly used as visual plans for embodied agents, but optimizing purely for visual plausibility often fails to capture the fragile physical manifold of real-world interactions. Even minor physics-violating errors at the interaction boundary, such as interpenetration or premature object motion, can completely invalidate the inferred timing and pose needed for downstream execution. Because standard supervised fine-tuning lacks the direct pressure to penalize these localized failures, we introduce *AgiBot-PhysPref*. This rigorously curated 10,000-sample preference dataset isolates condition-matched physics-violations, turning the generator’s own failure distribution into a foundational signal for physical consistency. Building upon this, we propose *RobotAPO*, an adversarial physics preference optimization framework operating in the continuous flow-matching denoising space. To prevent the policy from merely memorizing static curated failures, *RobotAPO* employs a lightweight adversarial counterfactual proposer that learns a condition-dependent, physical-failure-biased direction in denoising space. This encourages the model to explore and better respect the physical interaction boundary, all while maintaining a pure prompt-and-reference inference interface without requiring external structural conditioning. Comprehensive evaluations demonstrate that explicitly correcting these localized physics-violations improves downstream robot execution from generated videos. On held-out AgiBot conditions, *RobotAPO* outperforms the strongest controlled internal baseline in physical consistency by 6.8% hard score and 10.0% soft score. Crucially, in real-robot replay, it translates these physical-consistency gains into a 37.4% relative improvement in task success over the strongest controlled internal baseline.

1. Introduction

Modern diffusion and image-conditioned video priors have made video synthesis increasingly practical (Blattmann et al., 2023a,b). Large text/image-to-video generators further broaden prompt- and reference-conditioned generation (Bao et al., 2024; Hong et al., 2022; Yang et al., 2024), with recent large-scale video backbones pushing this direction further (Kong et al., 2024; Team et al., 2025b). Video generation has therefore become a practical interface for embodied intelligence. Generated videos can serve as task-conditioned visual plans and predictive rollouts for robot learning (Black et al., 2023; Du et al., 2023; Zhou et al., 2024). They also support embodied world modeling and policy learning (Feng et al., 2025; Hu et al., 2024; Wen et al., 2024). Related action-learning systems exploit generated or actionless videos as supervision (Ko et al., 2023; Wu et al., 2023). Synthetic demonstrations and embodied data generation provide another route for scaling robot learning (Bharadhwaj et al., 2024; Chi et al., 2025; Jang et al., 2025; Li et al., 2025). Recent generative transfer pipelines further synthesize robot demonstrations or adapt experience across domains (Liu et al., 2025b; Team et al., 2025a; Wang et al., 2025).

In robotic manipulation, however, visual plausibility is only a partial criterion. While supervised fine-tuning (SFT) effectively aligns the model to the broad *visual manifold* of robot appearance and workspace layouts, it often fails to capture the narrow and fragile *physical manifold* governing contact, grasping, and object-state transitions. A useful generated video must strictly preserve these underlying physical laws. A single-frame

physical inconsistency, such as a gripper closing prematurely or an object moving before contact, can instantly invalidate the inferred timing and pose needed for downstream execution. Because SFT relies on demonstration reconstruction, it receives no direct training pressure to delineate these critical interaction boundaries from harmless perceptual variations. Consequently, SFT generators continue to produce localized physics-violating errors (e.g., interpenetration, unstable contact) that severely limit their reliability as embodied visual plans.

Preference optimization provides a natural route to contrastive supervision (Huaisheng et al., 2025; Liu et al., 2025a; Rafailov et al., 2023; Wallace et al., 2024). This motivates a preference signal that is both condition-matched and failure-aware: under the same instruction and first-frame context, a physically grounded manipulation must be strictly preferred over a generated rollout that violates physical causality. To this end, we introduce *AgiBot-PhysPref*, a preference dataset built from 40,000 SFT generator rollouts across 200 AgiBot task IDs (Bu et al., 2025). Following VLM-based screening and rigorous double human annotation, it provides 10,000 curated pairs isolating local physics-violations. Because the positive and negative videos share all visual and semantic conditions, this dataset turns the generator’s own failure distribution into a foundational signal for physical consistency.

Building upon this insight, we propose *RobotAPO*, an adversarial physics preference optimization framework in the continuous flow-matching denoising space. To prevent the policy from merely memorizing static failures, *RobotAPO* introduces a lightweight *adversarial counterfactual proposer*. Rather than serving as a standalone discriminator, this proposer learns a condition-dependent, physical-failure-biased direction in the flow-matching denoising space. It constructs hard negatives that explore the interaction boundary while seeking to preserve the global scene condition. This mechanism improves preference learning around physics-violating interactions without requiring external structural conditioning (e.g., depth, masks, or physical simulators). At inference, *RobotAPO* remains a pure, unconstrained prompt-and-reference-image generator.

We evaluate *RobotAPO* through automatic video metrics, real-robot replay, and failure-focused qualitative analysis. Together, these evaluations confirm that correcting localized physics-violations narrows the gap between generated videos and downstream robot execution. On held-out AgiBot conditions, *RobotAPO* outperforms the strongest controlled internal baseline in physical consistency. Crucially, in AnyPos-driven real-robot replay, it translates these physical-consistency gains into a substantial 31.4 absolute percentage point improvement in task success over the SFT baseline.

Our main contributions are formulated as follows:

- We identify that the critical bottleneck in generative robotic plans lies in physics-violating interaction boundaries. We introduce *AgiBot-PhysPref*, a 10,000-sample condition-matched preference dataset explicitly designed to isolate and penalize these local physical failures.
- We propose *RobotAPO*, a novel denoising-space preference optimization framework. By introducing an adversarial counterfactual proposer that learns a physical-failure-biased denoising direction, *RobotAPO* improves physical compliance along the interaction boundary while maintaining a pure prompt-and-reference inference interface.
- We demonstrate that explicitly penalizing physics violations translates directly to downstream embodied usefulness. Alongside superior automatic physical-consistency metrics, *RobotAPO* effectively bridges the video-to-action gap, achieving a 64.7% real-robot task success rate with a 37.4% relative improvement over the strongest controlled internal baseline.

2. Related Work

Manipulation video generation. Open-domain models provide the generative substrate for robot video synthesis, which are widely used as visual plans (Black et al., 2023; Du et al., 2023; Tan et al., 2025; Zhou et al., 2024), embodied world models (Feng et al., 2025; Hu et al., 2024; Wen et al., 2024), and for demonstration transfer (Dong et al., 2025; Liu et al., 2025b; Team et al., 2025a). Most systems rely on reconstruction-based

or prediction-based training, effectively capturing the broad visual manifold of scene appearance and coarse dynamics. However, for embodied execution, visual plausibility is insufficient, and generated plans must strictly respect physical interaction boundaries (e.g., precise contact timing and grasp maintenance). While recent VLM-based evaluators provide useful diagnostics for general visual and common-sense consistency (He et al., 2024, 2025), real-world execution remains the ultimate test for these localized physical requirements.

Alignment and preference optimization. Preference optimization aligns generative models with criteria beyond standard reconstruction. Diffusion preference methods implement comparisons in likelihood, score, or denoising spaces (Black et al., 2024; Huaisheng et al., 2025; Prabhudesai et al., 2023; Wallace et al., 2024), while recent video frameworks optimize visual quality, semantic alignment, and global plausibility (Cheng et al., 2025a,b; Hu et al., 2025; Liu et al., 2025a). Concurrently, robot learning heavily exploits self-generated rollouts for data filtering and augmentation (Cheang et al., 2024; Jang et al., 2025; Kim et al., 2024; O’Neill et al., 2024). Despite these advances, existing objectives rarely target the highly localized, action-critical physics-violations inherent in robotic manipulation. To bridge this gap, our work explicitly aligns the generation process with physical causality at the interaction boundary rather than generic visual preferences.

3. Method

3.1. Problem and Overview

Throughout the paper, backbone refers only to the Wan2.2-TI2V architecture. We refer to the robot-data-specialized checkpoint before adding the *RobotAPO* LoRA adapters as the SFT generator. This same checkpoint supplies the frozen SFT-centering scores during optimization and serves as the SFT baseline in experiments. While the SFT generator captures the broad visual manifold of robotic workspaces, its rollouts frequently deviate from the underlying physical manifold. We post-train this generator to explicitly respect physical interaction boundaries, ensuring its samples are both visually plausible and useful for execution-oriented visual planning.

A manipulation condition c contains a task instruction and a first-frame visual context. Each training example compares two videos under the same condition. One is a real successful demonstration that lies safely within the physical manifold. The other is a generated rollout that falls off this manifold, exhibiting a localized physics-violating error. We write each condition-matched preference sample as (c, x^+, x^-, e^-) with $x^+ \succ x^-$. Here, x^+ is the corresponding real AgiBot demonstration under c . x^- is a rollout from the SFT generator that exhibits a visible local physical failure under the same c . e^- is the primary failure label, chosen as the most severe and visually salient error when multiple failures are present. Because x^+ and x^- share instruction, scene, embodiment, and visual condition, the binary preference $x^+ \succ x^-$ reduces scene- and semantic-level variation and focuses supervision on physical interaction validity. The remaining real-versus-generated source asymmetry is examined with a source-matched control in Sec. A.1.

RobotAPO (Fig. 1) attaches two low-rank adaptation (LoRA) modules to the frozen Wan2.2 text-image-to-video (TI2V) SFT generator (Team et al., 2025b): a trainable policy generator f_π and a training-only adversarial proposer f_a . During optimization, the frozen SFT generator serves as a stop-gradient centering model for DPO-style scoring. While preference pairs penalize known condition-matched failures, f_a explores nearby invalid interactions omitted by the static dataset. This encourages f_π to optimize against condition-preserving negatives. At inference, all auxiliary branches are discarded and only f_π is deployed. It retains the standard prompt-and-reference inference interface, requiring no external structural conditioning (e.g., action, trajectory, masks, or depth).

3.2. AgiBot-PhysPref Construction

Failure taxonomy. We adopt a unified five-category taxonomy of visible local physical failures: interpenetration, entity consistency, contact causality, unsupported floating after release, and invalid interaction/other. Inter-

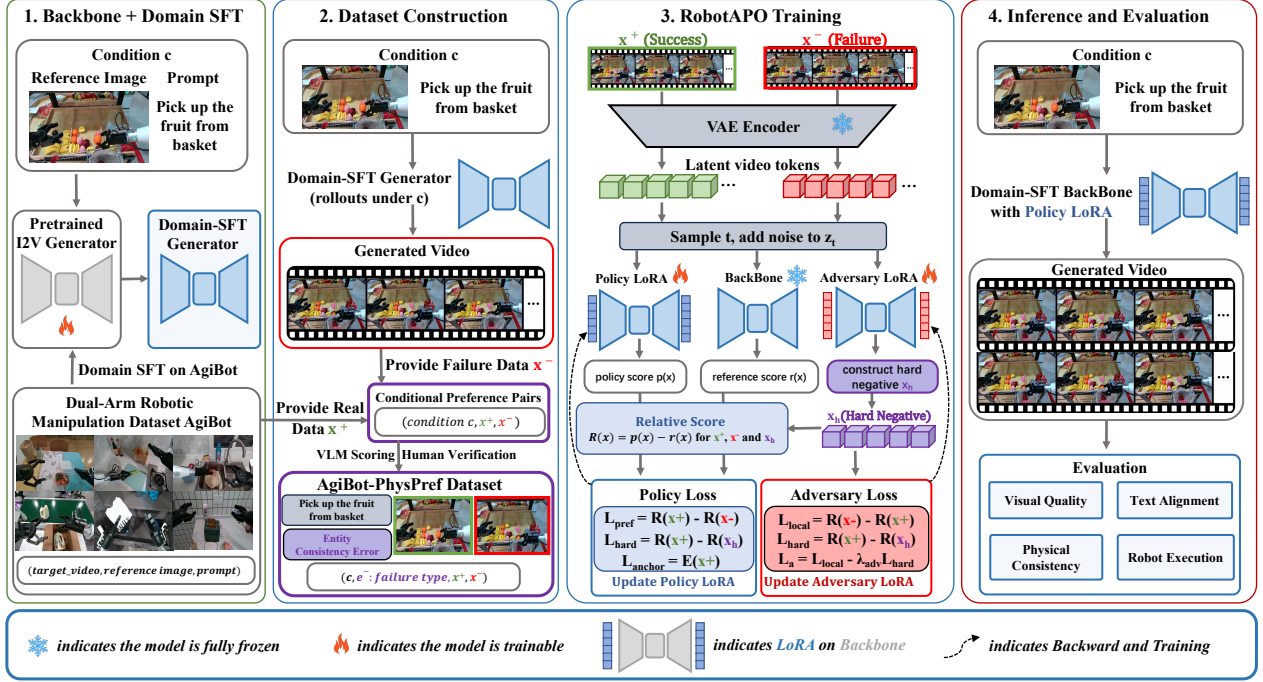


Figure 1: Overview of *RobotAPO*. SFT rollouts are screened and annotated to build the condition-matched *AgiBot-PhysPref*. During post-training, an adversarial proposer learns a physical-failure-biased denoising direction to probe interaction boundaries, encouraging the policy to better respect the physical manifold. At inference, only the policy generator is deployed, maintaining a standard prompt-and-reference interface without any external structural conditioning. Displayed scores are schematic, while exact denoising rewards and objectives are defined in Sec. 3.3.

penetration covers robot, object, or scene geometry passing through another entity. Entity consistency covers changes in object appearance, shape, identity, or existence over time. Contact causality covers object motion or state transitions without valid contact or manipulation, including premature motion before effective contact. Unsupported floating covers objects that remain suspended after losing support, while invalid interaction/other covers missed contacts, grasping air, or task-condition violations. The same taxonomy is used for human annotation, qualitative analysis, and the failure-focused figure. Full definitions are provided in Sec. B.5.

Construction pipeline. Starting from manipulation conditions in the *AgiBot* robot-manipulation benchmark (Bu et al., 2025), we roll out the SFT generator to obtain a candidate pool of 40,000 generated manipulation videos covering 200 task IDs over dual-arm scenarios. We then use Qwen3-VL-30B-Instruct as a quality-control screener. It ranks candidates by robot-interaction quality, 3D consistency, and instruction following. Annotators inspect candidates in descending screening order until 10,000 negatives are accepted. They review 12,935 candidates and exclude 2,935 videos that lack a visible physical failure or sufficient evidence for reliable labeling. Every reviewed candidate is independently assessed as a full video by two mutually blinded annotators, with disagreements sent to panel adjudication. The pre-adjudication seven-way judgments agree on 10,579/12,935 candidates (81.79%), with nominal Krippendorff’s $\alpha = 0.7832$. Every sample in the final set has a human-verified physical-failure label. When multiple visible failures are present, annotators record the primary label as the most severe and visually salient error. Each verified failed rollout is paired with the matching real *AgiBot* demonstration under the same condition, producing the curated 10,000-sample condition-matched preference set. The Qwen3-VL screening prompt, annotation protocol, and failure taxonomy are provided in Secs. B.1 to B.5.

We denote the resulting preference set by $\mathcal{D}_{\text{phy}} = \{(c_i, x_i^+, x_i^-, e_i^-)\}_{i=1}^N$ with $N = 10,000$, on which training only uses the binary relation $x_i^+ \succ x_i^-$. Holding the condition fixed prevents the model from picking up spurious

Table 1: *AgiBot-PhysPref* dataset statistics. The curated preference set isolates local physics violations via condition-matched pairs, human verification, and Qwen3-VL screening.

Property	Value
Generated candidates	40,000
Human-reviewed candidates	12,935
Final curated preference samples	10,000
Task IDs	200 dual-arm AgiBot scenarios
Screening model	Qwen3-VL-30B-Instruct
Final label source	Human double annotation with disagreement review
Pre-adjudication agreement	81.79%; Krippendorff’s $\alpha = 0.7832$
Preference unit	Condition-matched tuple (c, x^+, x^-, e^-)
Failure focus	Human-verified visible local physical failures
Failure labels	Interpenetration, entity consistency, contact causality, unsupported floating, invalid interaction/other

scene-level preferences and forces optimization to focus on grasp, contact, and object-motion quality. The primary failure annotations $\{e_i^-\}$ are kept for balancing the training mix across the five categories and for failure-focused qualitative analysis. They are metadata rather than multi-class rewards. Overlapping visible failures can be noted during review, but training stores the primary label.

3.3. Denoising-Space Physical-Failure Preference Learning

RobotAPO instantiates pairwise preference learning in the same flow-matching denoising space used to train the video generator. For a video x , let z_0 be its variational autoencoder (VAE) latent. For each training pair, we sample a flow-matching noise level σ and Gaussian noise ϵ . Under the rectified-flow schedule used in our Wan2.2 implementation, $z_\sigma(x) = (1 - \sigma)z_0(x) + \sigma\epsilon$ and $u_\sigma(x) = \epsilon - z_0(x)$, so a clean-latent estimate from a velocity prediction $\hat{u}$ is $z_\sigma - \sigma\hat{u}$. The task instruction and first-frame reference image are fed through the same image-to-video conditioning mechanism as the SFT generator. We denote the resulting condition by c . We package the denoising target as $y_x = (z_\sigma(x), u_\sigma(x))$ for a video x . Thus the preferred demonstration and curated failed rollout produce y_{x^+} and y_{x^-} , respectively. Counterfactual hard negatives introduced later use the same target format.

For a generator $b \in \{\pi, a, \text{SFT}\}$ with velocity prediction $\hat{u}_b(z, \sigma, c)$ and any denoising target $y = (z_y, u_y)$, define the denoising energy as

$$\mathcal{E}_b(y, \sigma, c) = \|\hat{u}_b(z_y, \sigma, c) - u_y\|_2^2. \quad (1)$$

In implementation this squared error is averaged over latent dimensions and minibatch samples. For readability in the loss formulas, we write $\mathcal{E}_b(y)$ as shorthand for $\mathcal{E}_b(y, \sigma, c)$ when σ and c are fixed within a training pair. The operator $\text{sg}(\cdot)$ denotes stop-gradient: its argument is used as a constant and receives no gradient. For either trainable branch $b \in \{\pi, a\}$, we define the same SFT-centered denoising reward

$$R_b(y) = -\mathcal{E}_b(y) + \text{sg}(\mathcal{E}_{\text{SFT}}(y)). \quad (2)$$

A larger $R_b(y)$ means that branch b assigns lower denoising error to y relative to the frozen SFT generator. In other words, R_π and R_a are the policy and adversary instances of the same score, with the same formula and different branch weights. The losses below compare reward differences between successful and negative targets, which keeps the DPO-style comparison local to the starting checkpoint while making the manipulation-specific negative targets the source of preference signal.

Curated physical-failure preference. We write $\text{sig}(\cdot)$ for the logistic sigmoid, kept distinct from the flow-matching noise level σ . The scalar $\gamma > 0$ is the preference temperature that scales the reward difference before

the sigmoid. For a condition-matched physical-failure pair (c, x^+, x^-) , the real successful video contributes y_{x^+} and the curated failed rollout contributes y_{x^-} . This keeps the pairwise preference form of DPO (Huaisheng et al., 2025; Rafailov et al., 2023; Wallace et al., 2024), but the comparison is explicitly grounded in flow-matching denoising energy and condition-matched manipulation failures. Both targets are evaluated under the same prompt, first-frame reference, noise level, and target construction, so the preference signal is aimed at local interaction errors rather than semantic shifts.

Preferred-sample anchor. To prevent the pairwise update from over-amplifying and damaging valid behavior of the SFT generator, we use a preferred-sample anchor and a LoRA drift penalty:

$$\mathcal{L}_{\text{anc}} = \mathcal{E}_\pi(y_{x^+}, \sigma, c), \quad \mathcal{L}_{\text{reg}} = \|\text{LoRA}(f_\pi)\|_2^2. \quad (3)$$

The anchor keeps the policy close to valid denoising behavior on successful trajectories, and the regularizer is a small weight penalty on the policy LoRA adapter.

3.4. Adversarial Counterfactual Hard-Negative Generation

Static curated failures become less informative once the policy learns to avoid obvious visual artifacts. To keep the negatives informative, *RobotAPO* introduces an adversarial counterfactual mechanism that constructs hard negatives near the physical interaction boundary.

Hard-negative construction. Given a successful video x^+ and its noisy latent z_σ^+ , we construct an adversarial counterfactual target by blending the policy and adversarial velocity predictions with a small mixing weight λ sampled from the range reported in Sec. B.6:

$$\hat{u}_h = \hat{u}_\pi(z_\sigma^+, \sigma, c) + \lambda [\hat{u}_a(z_\sigma^+, \sigma, c) - \hat{u}_\pi(z_\sigma^+, \sigma, c)]. \quad (4)$$

Here λ is only the hard-negative blending coefficient in latent denoising space. It is distinct from the adversary loss weight λ_{adv} introduced below. The counterfactual clean-latent estimate is $z_0^h = z_\sigma^+ - \sigma \hat{u}_h$. We write the hard-negative target as

$$y_h = h_a(y_{x^+}, \sigma, c, \lambda). \quad (5)$$

Concretely, $y_h = (z_\sigma^+, u_h)$ with $u_h \equiv \hat{u}_h$, whose implied clean latent under the same flow schedule is z_0^h . Because this perturbation starts from the successful noisy latent z_σ^+ , it is designed to preserve the condition while moving toward interactions that reduce the policy’s preference margin. The decoded comparison in Sec. A.2 evaluates whether the learned direction preferentially exposes local physical failures rather than generic corruption.

RobotAPO preference loss and policy objective. The preference term ranks the successful target above two manipulation-specific negatives: the curated failed rollout and the adversarial counterfactual target. We denote the negative target set for the current training example as

$$\mathcal{N}_{\text{neg}} = \{y_{x^-}, y_h\}. \quad (6)$$

$$\mathcal{L}_{\text{pref}} = \sum_{y^- \in \mathcal{N}_{\text{neg}}} -\log \text{sig}\left(\gamma(R_\pi(y_{x^+}) - R_\pi(y^-))\right). \quad (7)$$

The first summand uses the curated failed rollout y_{x^-} , and the second summand uses the counterfactual hard negative y_h . The full policy objective is then

$$\mathcal{L}_\pi = \mathcal{L}_{\text{pref}} + \mathcal{L}_{\text{anc}} + \mathcal{L}_{\text{reg}}, \quad (8)$$

Here $\mathcal{L}_{\text{anc}}$ and $\mathcal{L}_{\text{reg}}$ are defined in Eq. (3). During a policy update, only the LoRA parameters of f_π are updated, while f_a and the SFT centering model are frozen.

Adversary objective. The adversarial proposer is a generator-side module rather than a discriminator. It does not score decoded videos directly and is never decoded to pixels at inference. Instead, it combines an

inverted pairwise signal on the real success-vs-failure pair with a hard-negative pressure. During this update, the policy weights are frozen, but gradients still flow back through the adversary-induced hard target y_h :

$$\mathcal{L}_{\text{hard}}^{(a)} = -\log \text{sig}\left(\gamma(R_{\pi}(y_{x+}) - R_{\pi}(y_h))\right) \quad (9)$$

The policy term is evaluated with stop-gradient parameters, and the counterfactual target y_h remains differentiable with respect to f_a through Eqs. (4) and (5). The adversary-side local preference and full objective are

$$\mathcal{L}_{\text{local}}^{(a)} = -\log \text{sig}\left(\gamma(R_a(y_{x-}) - R_a(y_{x+}))\right) \quad (10)$$

$$\mathcal{L}_a = \mathcal{L}_{\text{local}}^{(a)} - \lambda_{\text{adv}} \mathcal{L}_{\text{hard}}^{(a)} \quad (11)$$

The first term uses the same reward form as Eq. (2), evaluated on the adversary branch, and makes the proposer prefer the failed target over the successful one. The second term increases the policy’s counterfactual loss, with λ_{adv} controlling the coupling strength.

Equations 8 and 11 establish a dynamic Min-Max game around the physical interaction boundary. The adversary searches for plausible velocity perturbations that reduce the policy’s preference margin, while the policy is trained to increase this margin. This boundary exploration is performed in the continuous denoising space and does not require external 3D physical simulators, depth maps, or mask constraints. Training begins with a policy-only warm-up and then alternates between policy and adversary phases, updating only the active branch while the SFT centering model remains frozen throughout. The λ sampling range, optimizer settings, and full schedule are reported in Sec. B.6.

4. Experiments

We design our experiments to validate whether explicitly optimizing for physical causality improves downstream robot execution from generated videos. We evaluate generated videos along three axes: 1) Automatic visual, alignment, and physical-consistency scoring, 2) Downstream embodied usefulness through real-robot pick-and-place execution, and 3) Local physical behavior through failure-focused qualitative analysis at the interaction boundary.

4.1. Experimental Setup

Backbone, data, and fair comparison. To strictly isolate the impact of training objectives, all internal methods share the Wan2.2-TI2V backbone (Team et al., 2025b), SFT initialization, condition-matched preference pairs, and inference settings (prompts, reference images, and samplers). The SFT baseline is evaluated without LoRA adapters, while Diffusion-DPO (Wallace et al., 2024) and RealDPO (Cheng et al., 2025a) serve as generic preference baselines. The full *RobotAPO* integrates anchor-preserving regularization and adversarial counterfactuals. We ablate these mechanisms via the *w/o Anchor/Reg* and *w/o CF/Adv* variants, respectively. To contextualize our 5B framework’s absolute performance, we also include LVP (Chen et al., 2025) (Wan2.1-I2V, 14B) as a high-capacity external scaling reference. Evaluation protocols for automatic scoring, robot replay, and qualitative inspection are detailed in their respective subsections.

4.2. Main Quantitative Comparison

For the AgiBot automatic comparison, we first train the SFT generator on curated dual-arm manipulation videos (Bu et al., 2025). All internal preference methods initialize from it, train on *AgiBot-PhysPref*, and are evaluated on disjoint held-out conditions with identical prompts, reference images, sampler settings, a fixed shared LoRA scale of 0.4, and random seeds. We use the official VideoScore2 evaluator (He et al., 2025) on 200 held-out task conditions with five samples per condition, for 1,000 videos per method. We report three

Table 2: Quantitative comparison on held-out AgiBot conditions. All internal methods are controlled 5B-parameter models. *RobotAPO* outperforms all controlled internal baselines; the external 14B LVP reference is shown in gray to contextualize absolute performance across a different model and training setup. Bold marks the best internal method.

Method	Size	Visual-H $\uparrow$	Align-H $\uparrow$	Phys-H $\uparrow$	Visual-S $\uparrow$	Align-S $\uparrow$	Phys-S $\uparrow$
LVP (Chen et al., 2025)	14B	3.647	3.787	3.473	3.030	3.252	3.363
Wan2.2-TI2V	5B	3.287	3.493	2.967	2.812	3.064	2.891
SFT baseline	5B	3.347	3.767	3.193	2.984	3.410	3.041
Diffusion-DPO (Wallace et al., 2024)	5B	3.340	3.760	3.253	3.120	3.484	3.172
RealDPO (Cheng et al., 2025a)	5B	3.347	3.760	3.173	3.047	3.435	3.008
<i>RobotAPO</i> (Ours)	5B	3.553	3.927	3.473	3.564	3.939	3.489

VideoScore2 (He et al., 2025) dimensions: visual quality, text-to-video alignment, and physical/common-sense consistency. For each dimension, **H** denotes the parsed hard integer score and **S** denotes the logit-based soft score, both on a 1–5 scale.

Tab. 2 reports official VideoScore2 scores on the held-out evaluation split. The gap between Wan2.2-TI2V and the SFT baseline confirms that robot-domain supervised fine-tuning is necessary to adapt the foundation prior to manipulation scenes. Compared with the SFT baseline, *RobotAPO* raises the hard scores for visual quality, alignment, and physical consistency to 3.553, 3.927, and 3.473, respectively. *RobotAPO* also improves the soft scores by 0.580 for visual quality, 0.529 for alignment, and 0.448 for physical consistency.

Among all internal Wan2.2-TI2V methods, *RobotAPO* sets the best mark on every hard and soft metric. Diffusion-DPO improves physical consistency over the SFT baseline but leaves the hard visual and alignment scores nearly unchanged. RealDPO yields smaller and less consistent gains, with physical scores occasionally falling below the SFT baseline. Together, these controlled comparisons suggest that generic preference objectives struggle to isolate the fragile physical manifold. The failure-aware supervision and adversarial hard-negative design of *RobotAPO* are essential for this setting. Crucially, visual quality also improves concurrently, indicating that strictly enforcing physical interaction boundaries does not trade off against global video plausibility.

The 14B external baseline, LVP, attains the highest hard visual-quality score and ties *RobotAPO* on Phys-H, reflecting the representational strength of its larger parameter scale. At roughly one-third the parameter scale, *RobotAPO* has the same observed Phys-H and higher observed soft scores and Align-H in this evaluation. Since LVP is an external reference rather than a matched-training baseline, these values contextualize performance but do not isolate the effect of the *RobotAPO* objective.

4.3. Robot Execution from Generated Videos

While automatic metrics capture perceptual plausibility, the ultimate test of physical manifold adherence is downstream execution. We run a separate real-robot replay evaluation on an AgileX platform and video domain. SFT, Diffusion-DPO, RealDPO, and *RobotAPO* start from their respective AgiBot-trained checkpoints and undergo target-domain adaptation on the same 400 self-collected AgileX videos, with matched optimization schedules and trainable-parameter budgets. Thus, the held-out AgiBot experiment evaluates *AgiBot-PhysPref* directly, while the AgileX study evaluates limited-data cross-platform adaptation from the corresponding AgiBot-trained checkpoints. LVP is evaluated directly from its released 14B checkpoint to provide a high-capacity, out-of-the-box reference point. All generated videos are passed through the same frozen AnyPos video-to-action pipeline (Tan et al., 2025) and controller. The recovered actions are replayed on 51 matched trials (17 task-scene configurations, three repeats each). A total of 51 videos per method and 255 videos overall are retained and executed without filtering, regeneration, trajectory correction, or manual intervention; every video yields a

Table 3: Real-robot pick-and-place execution success rates via AnyPos action extraction. *RobotAPO* improves downstream robot execution from generated videos, raising absolute task success by 31.4 percentage points over the SFT baseline. LVP is an external, unmatched reference. (Task success 95% CI uses the Wilson score interval over 51 trials).

Method	Size	Grasp Success (%) $\uparrow$	Cond. Place Success (%) $\uparrow$	Task Success (%) $\uparrow$	Task Success 95% CI
LVP	14B	72.5	86.5	62.7	[49.0, 74.7]
SFT baseline	5B	56.9	58.6	33.3	[22.0, 47.0]
Diffusion-DPO	5B	64.7	69.7	45.1	[32.3, 58.6]
RealDPO	5B	66.7	70.6	47.1	[34.1, 60.5]
<i>RobotAPO</i> (Ours)	5B	78.4	82.5	64.7	[51.0, 76.4]

Table 4: Module ablation of *RobotAPO*. Removing the adversarial counterfactual proposer (w/o CF/Adv) reduces real-robot task success from 64.7% to 49.0%, supporting the contribution of dynamic boundary exploration in this setting.

Method	Visual-S $\uparrow$	Align-S $\uparrow$	Phys-S $\uparrow$	Grasp Success (%) $\uparrow$	Task Success (%) $\uparrow$
SFT baseline	2.984	3.410	3.041	56.9	33.3
<i>RobotAPO</i> w/o CF/Adv	3.194	3.614	3.134	68.6	49.0
<i>RobotAPO</i> w/o Anchor/Reg	3.000	3.326	3.079	60.8	35.3
<i>RobotAPO</i> (Ours)	3.564	3.939	3.489	78.4	64.7

finite controller-accepted trajectory. A trial is successful only if the robot completes the full pick-and-place instruction. We report a Wilson 95% confidence interval for task success.

Tab. 3 shows that *RobotAPO* attains the highest observed task success rate (64.7%, 33/51). It improves over the SFT baseline by 31.4 absolute percentage points under the matched adaptation and AnyPos extraction pipeline; the configuration-clustered paired 95% CI is [15.7, 49.0] percentage points. The 14B external baseline, LVP, reaches 62.7% observed task success. The paired *RobotAPO*–LVP difference is +2.0 percentage points with a 95% CI of [−9.8, 13.7], so the two systems attain similar observed success under the current trial budget rather than establishing a significant advantage over LVP. These results evaluate the end-to-end combination of each generator with the frozen AnyPos/controller interface.

4.4. Ablation Study

Module ablation. We ablate the two main design axes introduced by *RobotAPO*: the counterfactual adversarial hard-negative mechanism (w/o CF/Adv) and the preferred-sample anchor with regularization (w/o Anchor/Reg). Tab. 4 provides the most critical insight into the system’s performance. Static curated failures alone (w/o CF/Adv) are sufficient to improve generic VideoScore2 metrics (Visual-S rises to 3.194, Align-S to 3.614). However, without the dynamic adversarial boundary exploration, Task Success drops precipitously by 15.7 absolute percentage points (from 64.7% to 49.0%). This contrast supports our core hypothesis: while static DPO can align models to broad visual and semantic preferences, the learned physical-failure-biased direction provides additional pressure around interaction failures that matter for downstream execution. Furthermore, removing the anchor and regularization (w/o Anchor/Reg) collapses both alignment and physical-consistency scores, validating their role in preventing the policy from severely deviating from the valid denoising manifold established during SFT.

Sensitivity to λ_{adv} . We examine the adversarial coupling coefficient λ_{adv} in Tab. 5, which controls the intensity

Table 5: Sensitivity study of the adversarial coupling coefficient λ_{adv} . This coefficient controls the strength of the adversarial counterfactual term that perturbs the local physical interaction boundary during training. The dagger $\dagger$ marks the default setting used in *RobotAPO*.

λ_{adv}	Visual-H $\uparrow$	Align-H $\uparrow$	Phys-H $\uparrow$	Visual-S $\uparrow$	Align-S $\uparrow$	Phys-S $\uparrow$
0.00	3.407	3.793	3.300	3.284	3.704	3.270
0.02	3.460	3.853	3.373	3.392	3.784	3.351
0.05 $\dagger$	3.553	3.927	3.473	3.564	3.939	3.489
0.10	3.433	3.827	3.420	3.365	3.758	3.421

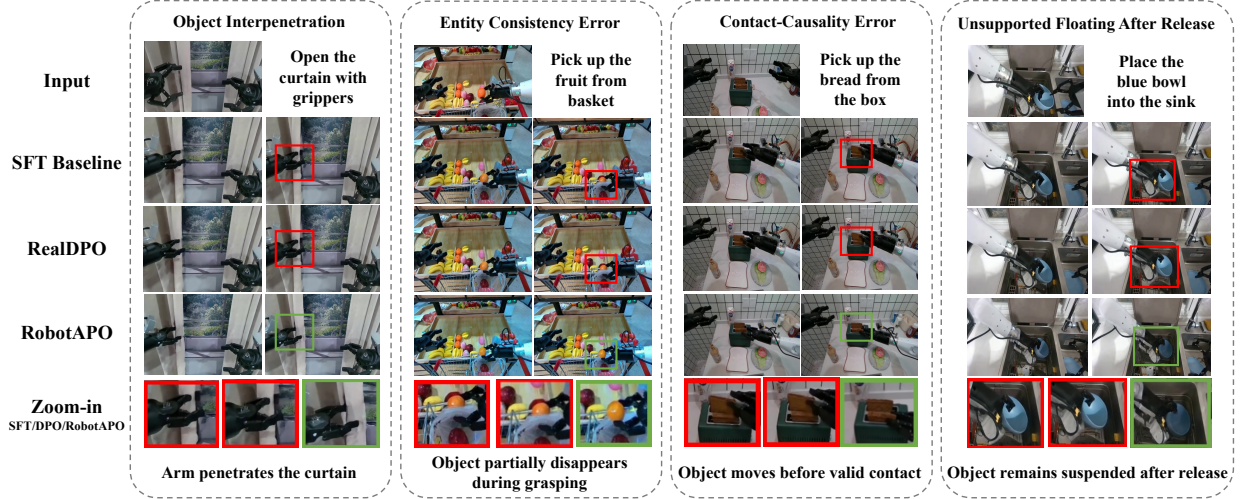


Figure 2: Failure-focused qualitative comparison on held-out conditions. For each task condition, all methods generate videos with the same frame length and inference settings. We visualize frames aligned to the same action-critical phase. Red boxes highlight physics-violating phenomena (e.g., interpenetration, premature object motion) frequently produced by baselines at action-critical moments. Green boxes show that *RobotAPO* reduces these local interaction artifacts and better preserves gripper–object geometry and contact causality. Zoom-ins isolate the key physical interactions.

of the counterfactual physical boundary perturbation. A weak adversarial coupling ($\lambda_{\text{adv}} = 0.02$) fails to provide sufficient pressure to explore the interaction boundary, yielding marginal gains. Conversely, an overly aggressive coupling ($\lambda_{\text{adv}} = 0.10$) disrupts the underlying visual manifold, harming both alignment and physical scores. The default setting (0.05) best balances aggressive adversarial boundary exploration with the structural stability of the visual manifold.

4.5. Failure-Focused Qualitative Comparison

Fig. 2 compares generated rollouts organized by our five-category failure taxonomy. Rather than presenting generic success cases, this analysis isolates critical interaction boundaries. While baselines may maintain high overall visual quality, they frequently show physical-causality errors at action-critical moments (e.g., interpenetrating the curtain, or the object shifting prematurely before the gripper closes). *RobotAPO* reduces these local interaction artifacts and better preserves gripper–object geometry, contact timing, and unsupported post-release physics. These qualitative boundary corrections are consistent with the gains in real-robot grasp and task success rates.

5. Conclusion

We presented *RobotAPO*, an adversarial preference optimization framework for improving physical consistency in robotic video generation. Leveraging our condition-matched *AgiBot-PhysPref*, *RobotAPO* optimizes physical preferences directly within the continuous flow-matching denoising space. To avoid memorizing static artifacts, an adversarial proposer learns a condition-dependent, physical-failure-biased denoising direction. This endogenous boundary exploration improves behavior around physical interaction failures without relying on external structural constraints. Real-robot evaluations show that reducing these localized physics violations improves end-to-end video-to-action execution under a fixed action-extraction interface. Future work will scale this paradigm and integrate closed-loop execution signals to further align generative visual plans with reliable physical control.

References

- Fan Bao, Chendong Xiang, Gang Yue, Guande He, Hongzhou Zhu, Kaiwen Zheng, Min Zhao, Shilong Liu, Yaole Wang, and Jun Zhu. Vidu: A highly consistent, dynamic and skilled text-to-video generator with diffusion models. *arXiv preprint arXiv:2405.04233*, 2024. 1
- Homanga Bharadhwaj, Debidatta Dwibedi, Abhinav Gupta, Shubham Tulsiani, Carl Doersch, Ted Xiao, Dhruv Shah, Fei Xia, Dorsa Sadigh, and Sean Kirmani. Gen2act: Human video generation in novel scenarios enables generalizable robot manipulation. *arXiv preprint arXiv:2409.16283*, 2024. 1
- Kevin Black, Mitsuhiko Nakamoto, Pranav Atreya, Homer Walke, Chelsea Finn, Aviral Kumar, and Sergey Levine. Zero-shot robotic manipulation with pretrained image-editing diffusion models. *arXiv preprint arXiv:2310.10639*, 2023. 1, 2
- Kevin Black, Michael Janner, Yilun Du, Ilya Kostrikov, and Sergey Levine. Training diffusion models with reinforcement learning and human feedback. *ICLR*, 2024. 3
- Andreas Blattmann, Tim Dockhorn, Sumith Kulal, Daniel Mendelevitch, Maciej Kilian, Dominik Lorenz, Yam Levi, Zion English, Vikram Voleti, Adam Letts, et al. Stable video diffusion: Scaling latent video diffusion models to large datasets. *arXiv preprint arXiv:2311.15127*, 2023a. 1
- Andreas Blattmann, Robin Rombach, Huan Ling, Tim Dockhorn, Seung Wook Kim, Sanja Fidler, and Karsten Kreis. Align your latents: High-resolution video synthesis with latent diffusion models. In *CVPR*, 2023b. 1
- Qingwen Bu, Jisong Cai, Li Chen, Xiuqi Cui, Yan Ding, Siyuan Feng, Shenyuan Gao, Xindong He, Xuan Hu, Xu Huang, et al. Agibot world colosseum: A large-scale manipulation platform for scalable and intelligent embodied systems. *arXiv preprint arXiv:2503.06669*, 2025. 2, 4, 7
- Chi-Lam Cheang, Guangzeng Chen, Ya Jing, Tao Kong, Hang Li, Yifeng Li, Yuxiao Liu, Hongtao Wu, Jiafeng Xu, Yichu Yang, et al. Gr-2: A generative video-language-action model with web-scale knowledge for robot manipulation. *arXiv preprint arXiv:2410.06158*, 2024. 3
- Boyuan Chen, Tianyuan Zhang, Haoran Geng, Kiwhan Song, Caiyi Zhang, Peihao Li, William T Freeman, Jitendra Malik, Pieter Abbeel, Russ Tedrake, et al. Large video planner enables generalizable robot control. *arXiv preprint arXiv:2512.15840*, 2025. 7, 8
- Guo Cheng, Danni Yang, Ziqi Huang, Jianlou Si, Chenyang Si, and Ziwei Liu. Realdpo: Real or not real, that is the preference. *arXiv preprint arXiv:2510.14955*, 2025a. 3, 7, 8
- Yuan Cheng et al. Discriminator-free preference optimization for video diffusion models. *arXiv preprint arXiv:2504.08542*, 2025b. 3

- Xiaowei Chi, Peidong Jia, Chun-Kai Fan, Xiaozhu Ju, Weishi Mi, Kevin Zhang, Zhiyuan Qin, Wanxin Tian, Kuangzhi Ge, Hao Li, et al. Wow: Towards a world omniscient world model through embodied interaction. *arXiv preprint arXiv:2509.22642*, 2025. [1](#)
- Zhehao Dong, Xiaofeng Wang, Zheng Zhu, Yirui Wang, Yang Wang, Yukun Zhou, Boyuan Wang, Chaojun Ni, Runqi Ouyang, Wenkang Qin, et al. Emma: Generalizing real-world robot manipulation via generative visual transfer. *arXiv preprint arXiv:2509.22407*, 2025. [2](#)
- Yilun Du, Sherry Yang, Bo Dai, Hanjun Dai, Ofir Nachum, Josh Tenenbaum, Dale Schuurmans, and Pieter Abbeel. Learning universal policies via text-guided video generation. *Adv. Neural Inform. Process. Syst.*, 2023. [1](#), [2](#)
- Yao Feng, Hengkai Tan, Xinyi Mao, Guodong Liu, Shuhe Huang, Chendong Xiang, Hang Su, and Jun Zhu. Vidar: Embodied video diffusion model for generalist bimanual manipulation. *arXiv preprint arXiv:2507.12898*, 2025. [1](#), [2](#)
- Xuan He, Dongfu Jiang, Ping Nie, et al. Videoscore: Building automatic metrics to simulate fine-grained human feedback for video generation. *arXiv preprint arXiv:2406.15252*, 2024. [3](#)
- Xuan He, Dongfu Jiang, Ping Nie, Minghao Liu, Zhengxuan Jiang, Mingyi Su, Wentao Ma, Junru Lin, Chun Ye, Yi Lu, et al. Videoscore2: Think before you score in generative video evaluation. *arXiv preprint arXiv:2509.22799*, 2025. [3](#), [7](#), [8](#), [18](#)
- Wenyi Hong, Ming Ding, Wendi Zheng, Xinghan Liu, and Jie Tang. Cogvideo: Large-scale pretraining for text-to-video generation via transformers. *arXiv preprint arXiv:2205.15868*, 2022. [1](#)
- Yucheng Hu, Yanjiang Guo, Pengchao Wang, Xiaoyu Chen, Yen-Jen Wang, Jianke Zhang, Koushil Sreenath, Chaochao Lu, and Jianyu Chen. Video prediction policy: A generalist robot policy with predictive visual representations. *arXiv preprint arXiv:2412.14803*, 2024. [1](#), [2](#)
- Zhiqiang Hu et al. D-fusion: Direct preference optimization for video diffusion. *arXiv preprint arXiv:2505.22002*, 2025. [3](#)
- Zhu Huaisheng, Xiao Teng, and Honavar Vasant. Dspo: Direct score preference optimization for diffusion model alignment. *ICLR*, 2025. [2](#), [3](#), [6](#)
- Joel Jang, Seonghyeon Ye, Zongyu Lin, Jiannan Xiang, Johan Bjorck, Yu Fang, Fengyuan Hu, Spencer Huang, Kaushil Kundalia, Yen-Chen Lin, et al. Dreamgen: Unlocking generalization in robot learning through neural trajectories. *arXiv e-prints*, 2025. [1](#), [3](#)
- Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, et al. Openvla: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024. [3](#)
- Po-Chen Ko, Jiayuan Mao, Yilun Du, Shao-Hua Sun, and Joshua B Tenenbaum. Learning to act from actionless videos through dense correspondences. *arXiv preprint arXiv:2310.08576*, 2023. [1](#)
- Weijie Kong, Qi Tian, Zijian Zhang, Rox Min, Zuozhuo Dai, Jin Zhou, Jiangfeng Xiong, Xin Li, Bo Wu, Jianwei Zhang, et al. Hunyuanvideo: A systematic framework for large video generative models. *arXiv preprint arXiv:2412.03603*, 2024. [1](#)
- Shuang Li, Yihuai Gao, Dorsa Sadigh, and Shuran Song. Unified video action model. *arXiv preprint arXiv:2503.00200*, 2025. [1](#)
- Jiachen Liu et al. Videodpo: Omni-preference alignment for video diffusion generation. *CVPR*, 2025a. [2](#), [3](#)

- Liu Liu, Xiaofeng Wang, Guosheng Zhao, Keyu Li, Wenkang Qin, Jiagang Zhu, Jiaxiong Qiu, Zheng Zhu, Guan Huang, and Zhizhong Su. Robottransfer: Controllable geometry-consistent video diffusion for manipulation policy transfer. *arXiv preprint arXiv:2505.23171*, 2025b. [1](#), [2](#)
- Abby O'Neill, Abdul Rehman, Abhiram Maddukuri, Abhishek Gupta, Abhishek Padalkar, Abraham Lee, Acorn Pooley, Agrim Gupta, Ajay Mandlekar, Ajinkya Jain, et al. Open x-embodiment: Robotic learning datasets and rt-x models. In *ICRA*, 2024. [3](#)
- Mihir Prabhudesai, Aniruddha Goyal, Deepak Pathak, and Katerina Fragkiadaki. Aligning text-to-image diffusion models with reward backpropagation. In *ICML Workshop*, 2023. [3](#)
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *arXiv preprint arXiv:2305.18290*, 2023. [2](#), [6](#)
- Hengkai Tan, Yao Feng, Xinyi Mao, Shuhe Huang, Guodong Liu, Zhongkai Hao, Hang Su, and Jun Zhu. Anypos: Automated task-agnostic actions for bimanual manipulation. *arXiv preprint arXiv:2507.12768*, 2025. [2](#), [8](#)
- GigaWorld Team, Angen Ye, Boyuan Wang, Chaojun Ni, Guan Huang, Guosheng Zhao, Haoyun Li, Jiagang Zhu, Kerui Li, Mengyuan Xu, et al. Gigaworld-0: World models as data engine to empower embodied ai. *arXiv preprint arXiv:2511.19861*, 2025a. [1](#), [2](#)
- Wan Team, Ang Wang, Baole Ai, Bin Wen, Chaojie Mao, Chen-Wei Xie, Di Chen, Feiwu Yu, Haiming Zhao, Jianxiao Yang, et al. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025b. [1](#), [3](#), [7](#)
- Bram Wallace, Meihua Dang, Rafael Rafailov, Linqi Zhou, Aaron Lou, Senthil Purushwalkam, Stefano Ermon, Caiming Xiong, Shafiq Joty, and Nikhil Naik. Diffusion model alignment using direct preference optimization. In *CVPR*, 2024. [2](#), [3](#), [6](#), [7](#), [8](#)
- Boyuan Wang, Xinpan Meng, Xiaofeng Wang, Zheng Zhu, Angen Ye, Yang Wang, Zhiqin Yang, Chaojun Ni, Guan Huang, and Xingang Wang. Embodiedreamer: Advancing real2sim2real transfer for policy training via embodied world modeling. *arXiv preprint arXiv:2507.05198*, 2025. [1](#)
- Youpeng Wen, Junfan Lin, Yi Zhu, Jianhua Han, Hang Xu, Shen Zhao, and Xiaodan Liang. Vidman: Exploiting implicit dynamics from video diffusion model for effective robot manipulation. *Adv. Neural Inform. Process. Syst.*, 2024. [1](#), [2](#)
- Hongtao Wu, Ya Jing, Chilam Cheang, Guangzeng Chen, Jiafeng Xu, Xinghang Li, Minghuan Liu, Hang Li, and Tao Kong. Unleashing large-scale video generative pre-training for visual robot manipulation. *arXiv preprint arXiv:2312.13139*, 2023. [1](#)
- Zhuoyi Yang, Jiayan Teng, Wendi Zheng, Ming Ding, Shiyu Huang, Jiazheng Xu, Yuanming Yang, Wenyi Hong, Xiaohan Zhang, Guanyu Feng, et al. Cogvideox: Text-to-video diffusion models with an expert transformer. *arXiv preprint arXiv:2408.06072*, 2024. [1](#), [15](#)
- Siyuan Zhou, Yilun Du, Jiaben Chen, Yandong Li, Dit-Yan Yeung, and Chuang Gan. Robodreamer: Learning compositional world models for robot imagination. *arXiv preprint arXiv:2404.12377*, 2024. [1](#), [2](#)

Table 6: Source-matched G+G control. Both sides of every training preference pair are generated by the same SFT model under the same condition.

Method	Visual-H $\uparrow$	Visual-S $\uparrow$	Phys-H $\uparrow$	Phys-S $\uparrow$
SFT	3.347	2.984	3.193	3.041
Diffusion-DPO	3.240	3.211	3.185	3.061
RealDPO	3.225	3.212	3.180	3.092
<i>RobotAPO</i> (Ours)	3.355	3.345	3.310	3.293

A. Additional Controlled Analyses

A.1. Source-Matched Generated-vs.-Generated Preferences

The original *AgiBot-PhysPref* construction pairs a real successful demonstration with a generated rollout containing a human-verified physical failure. Although the instruction and first-frame condition are matched, this construction does not remove the real-versus-generated source difference. We therefore conduct a source-matched generated-vs.-generated (G+G) control in which both videos in each preference pair are sampled from the same Wan2.2-TI2V SFT generator under the same *AgiBot* training condition.

For each condition, we generate ten rollouts with different seeds and rank them using a confidence-aware binary physical-error classifier trained only on *AgiBot-PhysPref* human labels, without VideoScore2 supervision. The least-flawed rollout is preferred and the most-flawed rollout is rejected. We construct 12,935 candidate pairs and remove the 2,935 pairs with the smallest classifier-margin gaps, retaining 10,000 G+G pairs. Independent human checking on 100 randomly sampled pairs yields an aggregate agreement of 84.5% with the classifier ordering.

Diffusion-DPO, RealDPO, and *RobotAPO* are retrained from the same SFT initialization with identical G+G data and training budgets. Evaluation follows the original protocol of 200 held-out conditions, five generations per condition, and 1,000 videos per method.

Under source-matched supervision, *RobotAPO* improves Phys-S over Diffusion-DPO by +0.232 (95% CI [0.132, 0.336]) and over RealDPO by +0.201 ([0.120, 0.289]). The corresponding Phys-H gains are +0.125 ([0.060, 0.190]) and +0.130 ([0.070, 0.190]). These condition-level paired bootstrap intervals resample the 200 conditions while retaining all five generations within each sampled condition. The persistent gains show that the advantage of *RobotAPO* cannot be explained solely by a direct source-identity cue.

A.2. Learned Direction vs. Matched Gaussian Perturbations

We test whether the adversarial residual captures failure-related structure beyond generic latent corruption. With the checkpoint and hyperparameters frozen, we randomly select 64 held-out conditions before decoding and generate one learned-adversarial and one equal-norm Gaussian-control video per condition. Each pair shares the prompt, reference frame, noisy latent, noise level, blending coefficient, decoder, and reference-latent mask. The Gaussian direction is normalized per sample to the ℓ_2 norm of the learned direction. All 128 videos are retained and assessed by two annotators blinded to variant identity, followed by blinded adjudication.

The learned direction produces condition-preserving local failures 1.85 times as often as the matched Gaussian control (95% CI [1.14, 3.38]; exact McNemar $p = 0.0266$). Among adversarial outputs exhibiting any failure or unusability, 75.0% are targeted local physical failures. Agreement before adjudication on local physical failure versus other is 89.8%, with Krippendorff’s $\alpha = 0.752$. These results support describing the mechanism as a physical-failure-biased denoising direction at the evaluated operating point, without implying explicit spatial localization.

Table 7: Decoded adversarial counterfactuals compared with equal-norm Gaussian controls over 64 paired held-out conditions.

Diagnostic	Adversarial	Gaussian	Paired effect [95% CI]
Condition-preserving local physical failure	37.5%	20.3%	+17.2 pp [3.1, 31.3]
Global corruption / condition change	9.4%	6.3%	+3.1 pp [−6.3, 12.5]
Unusable	3.1%	1.6%	+1.6 pp [−3.1, 6.3]

Table 8: CogVideoX-5B evaluation. Confidence intervals are condition-clustered paired bootstrap intervals for the RobotAPO Phys-S gain over each baseline.

Method	Visual-S	Phys-H	Phys-S	Gain [95% CI]
SFT	3.039 ± 0.199	2.980 ± 0.407	2.775 ± 0.626	+0.193 [0.088, 0.316]
Diffusion-DPO	3.009 ± 0.104	3.010 ± 0.415	2.824 ± 0.547	+0.144 [0.058, 0.245]
RealDPO	3.039 ± 0.198	2.970 ± 0.399	2.786 ± 0.607	+0.183 [0.081, 0.299]
<i>RobotAPO</i>	3.008 ± 0.176	3.020 ± 0.410	2.968 ± 0.294	–

A.3. Transfer to CogVideoX-5B

We repeat the complete controlled comparison on the architecturally distinct CogVideoX-5B backbone (Yang et al., 2024). Except for each backbone’s native denoising parameterization and sampler, all methods use the same preference pairs, split, conditioning inputs, training and LoRA budgets, held-out conditions, and evaluation protocol. The comparison uses 200 held-out conditions with five generations each.

All three paired intervals exclude zero. Together with the Wan2.2-TI2V results, this supports portability across the two evaluated video backbones while keeping the claimed scope explicit.

A.4. Uncertainty and Execution-Relevant Metric Analysis

For the Wan2.2-TI2V comparison, per-video Phys-H/Phys-S standard deviations are 0.813/0.837 for SFT, 0.866/0.893 for Diffusion-DPO, 0.782/0.825 for RealDPO, and 0.953/0.937 for *RobotAPO*. Condition-clustered paired bootstrap estimates for the RobotAPO Phys-S gain are +0.336 over SFT (95% CI [0.269, 0.401]), +0.243 over Diffusion-DPO ([0.178, 0.309]), and +0.370 over RealDPO ([0.306, 0.435]).

Across all 255 generated videos used for real-robot execution, the point-biserial correlation with binary full-task success is 0.775 ([0.721, 0.822]) for Phys-S, compared with 0.467 ([0.351, 0.568]) for Visual-S and 0.452 ([0.359, 0.527]) for Align-S. After jointly controlling for Visual-S and Align-S over the 200 held-out AgiBot conditions, *RobotAPO* retains adjusted Phys-S gains of +0.160 ([0.099, 0.227]), +0.101 ([0.042, 0.161]), and +0.223 ([0.161, 0.291]) over SFT, Diffusion-DPO, and RealDPO, respectively. We use Phys-S as a complementary indicator alongside complete-video inspection and robot execution rather than as a substitute for physical evaluation.

For real-robot task success, configuration-clustered paired differences are +31.4 percentage points over SFT ([15.7, 49.0]), +19.6 over Diffusion-DPO ([−2.0, 41.2]), +17.6 over RealDPO ([−2.0, 37.3]), and +2.0 over LVP-14B ([−9.8, 13.7]). Thus, the current trial budget supports the improvement over SFT; the other intervals include zero.

B. Dataset, Implementation, and Reproducibility Details

B.1. Dataset Documentation

To make *AgiBot-PhysPref* auditable and reproducible, each dataset version is tracked through a set of internal manifests. The condition manifest records the AgiBot video identifier, task instruction, visual-condition path, split assignment, and prompt hash. The generated-candidate manifest records the generator checkpoint, sampling seed, sampling hyperparameters, output path, and condition identifier for the full 40,000-video candidate pool. For the screening stage, we store the Qwen3-VL-30B-Instruct raw response, parsed quality scores, ranking position, parsing status, and filtering decision. For the human verification stage, we store the accept/reject decision, primary failure type, optional secondary failure notes, annotator identifier, and review status. The final preference-pair manifest contains the condition-matched tuples (c, x^+, x^-, e^-) used by training, validation, and evaluation, where x^+ is the real successful demonstration, x^- is the failed rollout, and e^- is the primary physical-failure label. The evaluation manifest records held-out prompts, generated outputs from each compared method, selected LoRA scale, official VideoScore2 outputs, and robot replay records when applicable.

B.2. Dataset Summary and Annotation Flow

AgiBot-PhysPref is built from a 40,000-video candidate pool generated by the SFT generator under 200 task-ID conditions from the AgiBot robot-manipulation benchmark. These conditions cover dual-arm manipulation scenarios. The dataset is used as a training preference set rather than as a standalone benchmark with an official test split. Evaluation is performed on disjoint held-out AgiBot task conditions, each paired with a real first-frame reference image and task instruction. The candidate pool is not used directly as training supervision. Instead, generated rollouts are first saved with their prompts, visual conditions, checkpoint identifiers, sampling seeds, and output paths. Qwen3-VL-30B-Instruct is then used as a quality-control screener to rank candidates by interaction quality, 3D consistency, and instruction following. Annotators inspect candidates in descending screening order until 10,000 negatives are accepted. This requires reviewing 12,935 candidates and excludes 2,935 videos without a visible physical failure or sufficient evidence for reliable labeling.

Each reviewed candidate is independently annotated as a full video by two mutually blinded annotators using the unified five-category failure taxonomy in [Sec. B.5](#), augmented with no-visible-failure and uncertain outcomes during curation. Annotators record the most severe and visually salient visible failure as the primary label. Their pre-adjudication seven-way judgments agree on 10,579/12,935 candidates (81.79%), with nominal Krippendorff’s $\alpha = 0.7832$. The remaining 2,356 candidates are sent to panel adjudication before the final preference manifest is constructed. Only human-verified condition-matched pairs are used as preference supervision, yielding the final 10,000 curated samples. Labels such as no significant physical issue and uncertain are retained for filtering and auditability, but they are not treated as physical-failure categories. Secondary notes are also retained for auditing, while the training objective uses only the binary preference relation and the primary label for balancing and analysis.

B.3. Qwen3-VL-30B-Instruct Screening Protocol

The Qwen3-VL stage is used only for candidate quality control during dataset construction. Official VideoScore2 is used separately for automatic evaluation in [Sec. 4.2](#). During screening, the implementation samples 16 uniformly spaced RGB frames per video, conditions Qwen3-VL-30B-Instruct on the task instruction, and uses deterministic decoding with `do_sample=False` and `max_new_tokens=400`. The model is asked to evaluate only visually observable evidence in the sampled frames and to return a valid JSON object with three integer scores and concise evidence-based reasons. The three scoring fields are `Interaction_Quality`, `Perspectivity`, and `Instruction_Following`. `Interaction_Quality` covers robot-object contact, penetration, friction, force transfer, grasping, lifting, and physically plausible manipulation. `Perspectivity` covers 3D consistency, depth

ordering, camera geometry, occlusion, scale stability, and floating artifacts. `Instruction_Following` compares the video against the task instruction and penalizes wrong objects, wrong actions, wrong embodiments, and human-hand hallucinations. Candidates are ranked by the mean of the three parsed scores, and the top 10,000 videos are sent to human annotation. The recorded metadata includes the Qwen3-VL model path or version, decoding parameters, raw response, parsed scores, parsing status, aggregate rank, and filtering decision for each candidate.

B.4. Human Verification Protocol

Human verification is used after VLM screening to ensure that every negative training sample contains a visible manipulation-related physical failure under the matched condition. Each candidate is independently labeled by two annotators, and disagreements are resolved through an additional review pass. Annotators reject or mark a candidate as uncertain when the video is corrupted, too unclear to judge, mismatched with the positive sample, unrelated to manipulation physics, or ambiguous enough that it cannot be reliably labeled as success or failure. Accepted negative samples receive one primary physical-failure label from the taxonomy in [Sec. B.5](#). When multiple visible failures are present, annotators choose the most severe and visually salient one and may record secondary notes for auditability. For AgiBot preference construction, only human-verified condition-matched pairs are used as training supervision.

B.5. Failure Type Taxonomy

[Tab. 9](#) restates the unified five-category taxonomy used for human annotation, training metadata, and failure-focused evaluation. The contact-causality category is intentionally inclusive: it covers both premature object motion before effective contact and action or state changes without effective contact.

Table 9: Unified five-category failure taxonomy used for *AgiBot-PhysPref* curation, training metadata, and failure-focused evaluation. The contact-causality category covers both premature object motion and action/state changes without effective contact.

Failure Type	Definition	Typical Evidence
Object interpenetration	Robot, object, or scene geometry passes through another entity.	Gripper penetrates object, or object sinks into table.
Entity consistency error	Robot or object identity changes in appearance, shape, or existence over time.	Object flickers, disappears, changes shape, or changes identity across frames.
Contact-causality error	Object motion or state transition is not causally supported by valid contact or manipulation, including premature motion before effective contact and action/state changes without effective contact.	Object moves before contact, state changes without effective contact, or gripper closes while the object slides without being grasped.
Unsupported floating after release	An object remains suspended after losing support.	Released object stays in mid-air or does not settle naturally.
Invalid interaction / other	No valid manipulation event under the task condition, or other visible failure not covered above.	Robot grasps air, never interacts with the target, or executes a misaligned motion that breaks the task.

The adjudicated primary-category distribution contains 2,072 interpenetration samples (20.72%), 1,632 entity-consistency samples (16.32%), 2,936 contact-causality samples (29.36%), 1,286 unsupported-floating samples (12.86%), and 2,074 invalid-interaction/other samples (20.74%). These mutually exclusive primary

labels sum to 10,000; secondary failure tags are retained for auditing, while training uses only the condition-matched binary preference.

B.6. Implementation Details

We start from the SFT Wan2.2-TI2V flow-matching generator. During *RobotAPO* post-training, the SFT generator weights are frozen and two LoRA adapters are attached to the attention projections to_q , to_k , to_v , and to_out . The policy adapter together with the SFT generator defines the deployed policy generator, while the adversary adapter is used only during post-training. At inference time, the SFT generator is loaded with only the policy adapter.

Each preference pair samples synchronized frame indices from the preferred and failed videos, applies shared resize-crop transforms, and normalizes frames to the model input range. Prompt embeddings are loaded from cache when available; otherwise they are computed with a frozen text encoder. First-frame reference conditioning follows the Wan2.2-TI2V latent-space conditioning interface through mask-guided composition, with expanded-timestep conditioning enabled when tighter temporal conditioning is required.

Optimization uses AdamW for both policy and adversary updates, BF16 mixed precision, DeepSpeed ZeRO-2, gradient accumulation, and gradient clipping. Training starts with a policy-only warm-up and then alternates policy and adversary phases. The adversary is warm-started from the policy LoRA weights at the first adversary phase. The training configuration records the pairwise temperature γ , the adversary coupling λ_{adv} , and the sampling range of the hard-negative mixing weight λ in Eq. (4). The regularizer $\mathcal{L}_{\text{reg}}$ is implemented as a small L2 weight-norm penalty on the trainable LoRA parameters of f_{π} .

B.7. Compute Resources

The main training runs used 8 NVIDIA RTX PRO 6000 GPUs for approximately 200 wall-clock hours, corresponding to about 1,600 GPU-hours. This estimate covers the main policy and adversary training runs reported in the paper. The Qwen3-VL screening, VideoScore2 evaluation, and robot replay stages are comparatively lighter-weight and are tracked separately in the experiment manifests.

B.8. Official VideoScore2 Evaluation Protocol

For automatic video scoring, each generated video and its task prompt are evaluated with the official VideoScore2 implementation (He et al., 2025). Following the released inference code, we report the three VideoScore2 dimensions: visual quality, text-to-video alignment, and physical/common-sense consistency. For each dimension, **H** denotes the parsed hard integer score and **S** denotes the official logit-based soft score, both on a 1–5 scale. The evaluation manifest records the video path, prompt, raw VideoScore2 output, parsed hard scores, soft scores, and any parsing errors.

B.9. Limitations and Broader Impacts

RobotAPO improves the physical reliability of robot manipulation video generation by explicitly penalizing visible interaction failures under matched task conditions. Its visually grounded preference objective sharpens the boundary between plausible rollouts and physically invalid ones, but it does not use explicit 3D geometry, force sensing, or rigid-body simulation priors. Thus, highly complex multi-object interactions, severe occlusion, deformable-object dynamics, and closed-loop execution feedback remain outside the technical focus of this study. The original *AgiBot-PhysPref* pairs a real successful demonstration with a generated failure, so it is condition-matched but not source-invariant. The source-matched G+G control in Sec. A.1 reduces this concern, but it does not change the construction of the original dataset. The demonstrated generalization scope is limited to two video backbones, held-out AgiBot conditions, and matched limited-data adaptation to the

AgileX platform; it does not establish zero-shot transfer across arbitrary embodiments or task families. In addition, *AgiBot-PhysPref* records one primary visible-failure label for each negative example; secondary artifacts are retained as annotation notes when useful, but are not modeled as separate rewards. Future work could incorporate richer state, depth, contact, or robot-execution signals to further improve visual-to-physical alignment.

Broadly, this work supports safer and more efficient research on embodied video generation by reducing physically implausible rollouts before robot replay or downstream execution. More physically consistent generated plans may reduce wasted real-world trials and make model failures easier to diagnose, but they should not be treated as verified execution plans without task-specific validation, safety checks, and human oversight. The data used in this work consists of robot manipulation videos and generated rollouts rather than personal or sensitive human data; any future release of data, annotations, or checkpoints will document intended use, known limitations, and applicable asset licenses.

B.10. Reproducibility Checklist

The end-to-end pipeline can be reproduced by following the same staged construction used in our experiments. First, one trains or loads the SFT Wan2.2-TI2V generator on AgiBot manipulation videos and generates the 40,000-video candidate pool for the 200 AgiBot benchmark task-ID conditions with recorded seeds and sampling parameters. Second, Qwen3-VL-30B-Instruct quality-control screening is run on the generated candidates, and the shortlisted videos are independently double-annotated before disagreement review. Third, the final train, validation, and held-out evaluation preference manifests are built with no condition leakage, yielding the 10,000 curated condition-matched preference samples. Fourth, prompt embeddings are precomputed and *RobotAPO* training is launched with policy warm-up followed by alternating policy and adversary phases. Finally, controlled internal baselines and ablations are trained under the same framework, and all internal checkpoints are evaluated with the official VideoScore2 protocol, a fixed shared LoRA scale of 0.4, the self-collected real-robot replay pipeline based on frozen AnyPos action extraction, and the failure-focused qualitative comparison protocol. LVP is evaluated only from its released checkpoint as an external embodied-video reference.